

Mathematical Analysis Lecture 21 积分学.

→ 第二换元法 $\int f(u) du = \int f(\varphi(x)) \varphi'(x) dx \mid x = \varphi^{-1}(u)$

e.g. $\int \sqrt{a^2 - x^2} dx$ 令 $x = a \sin t$ $\therefore \sqrt{a^2 - x^2} = \sqrt{a^2 - a^2 \sin^2 t} = a \cos t, t \in (-\frac{\pi}{2}, \frac{\pi}{2})$

$\therefore$ 原 = $\int a^2 \cos^2 t dt = a^2 \int \frac{1 + \cos 2t}{2} dt = \frac{a^2}{2} t + \frac{a^2 \sin 2t}{4} + C$ 代回 t : 原 = $\frac{a^2}{2} \arcsin \frac{x}{a} + \frac{1}{2} x \sqrt{a^2 - x^2} + C$

e.g. $\int \frac{dx}{\sqrt{x^2 + a^2}}$ 令 $x = a \tan t$ $\therefore \sqrt{x^2 + a^2} = a \sec t, dx = a \sec^2 t$

$\therefore$ 原 = $\int \sec t dt = \ln |\sec t + \tan t| + C_1 = \ln \left(\frac{\sqrt{x^2 + a^2} + a}{a} \right) + C_1 = \ln(x + \sqrt{x^2 + a^2}) + C$

e.g. $\int \frac{dx}{\sqrt{x^2 - a^2}}$ 令 $x = \sec t$ $\therefore$ 原 = $\int \sec t dt = \dots = \ln(x + \sqrt{x^2 - a^2}) + C \quad (x > a) = -\ln(-x + \sqrt{x^2 - a^2}) + C' = \ln(x + \sqrt{x^2 - a^2}) + C \quad (x < -a)$

Note: 对于 $\sqrt{x^2 \pm a^2}$, 除采用三角代换, 也可利用 $\cosh^2 t - \sinh^2 t = 1$, $\sinh x = \frac{e^x - e^{-x}}{2}$, $\cosh x = \frac{e^x + e^{-x}}{2}$ 但是反函数代换可能有麻烦.

$\int \sinh x dx = \cosh x + C, \int \cosh x dx = \sinh x + C$

e.g. $\int \frac{\sqrt{a^2 - x^2}}{x^4} dx$ 令 $x = \frac{1}{t}$, 则 $dx = -\frac{1}{t^2} dt$ 原 = $-\int \sqrt{a^2 t^2 - 1} \cdot \frac{1}{2a^2} d(a^2 t^2 - 1) \dots$

可以得: $-\frac{(a^2 - x^2)^{\frac{3}{2}}}{3a^2 x^3} + C$

三角代换也可以吧

e.g. $\int \frac{dx}{(x+1)^3 \sqrt{(x+1)^2 - 1}}$ 令 $x+1 = \frac{1}{t}$, 原 = $\int -\frac{t^2}{\sqrt{1-t^2}} dt = \int \frac{1-t^2-1}{\sqrt{1-t^2}} dt \dots$

→ 分部积分法: 由导数公式 $(uv)' = u'v + uv'$ $uv = \int u'v dx + \int uv' dx$

得到 $\int uv' dx = uv - \int u'v dx$ 即为分部积分法则 $\int u dv = uv - \int v du$

e.g. $\int x \cos x dx = x \sin x - \int \sin x dx = x \sin x + \cos x + C$

e.g. $\int x \ln x dx = \frac{1}{2} x^2 \ln x - \int \frac{1}{2} x dx = \frac{1}{2} x^2 \ln x - \frac{1}{4} x^2 + C$

e.g. $\int x \arctan x = \frac{1}{2} x^2 \arctan x - \frac{1}{2} \int \frac{x^2}{x^2+1} dx = \frac{1}{2} (x^2+1) \arctan x - \frac{1}{2} x + C$

e.g. $\int \ln x dx = x \ln x - \int x \cdot \frac{1}{x} dx = x \ln x - x + C$ (高中常用fm, 观察得)

e.g. $\int e^x \sin x dx = \sin x \cdot e^x - \int e^x \cos x dx = \sin x \cdot e^x - (e^x \cos x + \int e^x \sin x dx)$

$\therefore 2 \int e^x \sin x dx = e^x (\sin x - \cos x) + C' \therefore \int e^x \sin x = \frac{1}{2} e^x (\sin x - \cos x) + C$

e.g. $\int \arccos x dx = x \arccos x + \int \frac{x}{\sqrt{1-x^2}} dx = x \arccos x - \sqrt{1-x^2} + C$